

When a Decision Does Not Resolve

Non-resolution and stable uncertainty in AI judgement
systems

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Revision note – Version 1.1

This version refines the framing, abstract, introduction and interpretive discussion to improve clarity and accessibility. The core findings, dataset, method and conclusions remain unchanged.

Executive Summary

AI systems are often evaluated based on whether their outputs appear consistent, reasonable, or well explained. More recent approaches extend this by examining how those outputs behave when the same task is repeated.

Repeated evaluation usually focuses on variation. If a system changes its judgement across runs, this is interpreted as instability or sensitivity at the decision boundary.

This paper identifies a different behavioural pattern.

In some cases, the system does not vary in its judgement under repeated execution, but nor does it resolve the judgement into a definitive classification. Instead, it repeatedly returns the same indeterminate outcome.

Using repeated evaluations of job advertisements for potential age-related bias, the analysis shows that early-career roles containing signals such as “junior”, “graduate”, or “trainee” are consistently classified as “Unclear” across repeated runs.

This behaviour differs from instability. The system does not move between adjacent categories, but nor does it arrive at a decisive outcome. The judgement remains stable without resolving.

In practical settings, a stable but unresolved output may require escalation, abstention, or human review rather than being interpreted as evidence of reliable decision-making.

Abstract

Most approaches to evaluating AI systems focus on the quality of individual outputs or, more recently, on how those outputs vary when the same task is repeated. Variation across repeated runs is typically interpreted as instability in judgement.

This paper examines a different behavioural pattern: non-resolution. In some evaluative tasks, AI systems do not move between classifications under repeated execution, but instead remain fixed in an indeterminate state.

Using repeated evaluations of UK job advertisements for potential age-related bias, the analysis shows that certain early-career roles consistently produce “Unclear” classifications across multiple runs, with stable confidence and no evidence of threshold crossing or convergence toward more decisive outcomes.

This behaviour differs from instability. The system does not vary in its judgement, but nor does it resolve the judgement into a definitive classification. The findings suggest that non-resolution represents a distinct behavioural property of AI judgement systems, particularly in cases involving weak, mixed, or ambiguous signals.

The paper argues that stability alone is insufficient for evaluating AI reliability. A judgement may remain consistent while still failing to resolve into a usable decision. Behavioural evaluation therefore depends not only on whether outputs remain stable, but on whether they resolve at all.

Keywords

- AI evaluation
- LLM evaluation
- model reliability
- judgement stability
- behavioural audit
- decision boundaries
- persistent uncertainty

1. Introduction

Evaluation of AI systems is often based on the inspection of individual outputs. A response that appears reasonable or well explained is frequently taken as evidence that the system is behaving reliably.

This assumption becomes less certain when the same task is observed repeatedly. Earlier work has shown that AI judgements do not always remain the same under repeated evaluation. Where interpretation is required, variation emerges, particularly at decision boundaries.

In many cases, repeated evaluation reveals instability at the decision boundary, where classifications move between adjacent categories across runs. In other cases, however, the behaviour is different. The system does not vary in its judgement, but repeatedly returns the same indeterminate outcome without resolving into a decisive classification.

At decision boundaries, behaviour therefore does not follow a single pattern. Some judgements fluctuate across repeated runs, reflecting sensitivity to marginal differences in how signals are interpreted. Others remain fixed in an unresolved state despite repeated evaluation under identical conditions.

This paper examines that second pattern.

Building on earlier work on behavioural evaluation and judgement stability, the analysis focuses on non-resolution as a distinct behavioural property. Rather than asking whether a judgement remains the same, the question becomes whether the system is able to resolve the judgement at all.

Earlier work identified variation in judgement at decision boundaries, where classifications move between adjacent categories under identical conditions. The present analysis examines a different pattern, where judgements remain stable but indeterminate.

Taken together, these findings indicate that boundary cases do not produce a single type of behaviour. Some vary under repeated evaluation, while others remain stable without resolving. Both patterns are relevant to understanding how AI systems behave under ambiguity.

In operational settings, unresolved outputs may appear stable while still failing to provide a sufficiently decisive basis for action.

2. Research Context

This paper forms part of the *Agents at Work* research series, which examines how AI systems interpret age-coded language in recruitment text. The specific focus on early-career roles emerged from discussion, which prompted closer examination of this subset within the dataset.

Earlier work introduced the Behavioural Evaluation Framework, which proposes assessing AI systems through repeated execution, internal signals, and comparison across independent systems. This framework reframed evaluation as the observation of behavioural patterns rather than the inspection of individual outputs.

Subsequent analysis examined judgement stability under repeated execution. While most classifications remained consistent, a meaningful proportion varied under identical conditions, with variation concentrated at decision boundaries.

The present paper extends this line of enquiry.

Where earlier work focused on variation in judgement, this analysis examines cases where variation does not occur, but resolution is also absent. The focus is therefore not on instability, but on persistent indeterminacy within the classification process.

3. Method Overview

The analysis is based on repeated evaluations of job advertisements for potential age-related bias using a controlled instruction-tuned language model environment under consistent inference conditions.

Each advertisement was assessed multiple times using the same model, prompt, and input conditions. The classification task required assignment to one of four categories:

- Likely Biased
- Potentially Biased
- Unclear
- Unbiased

A subset of advertisements containing explicit early-career signals, such as “junior”, “graduate”, “trainee”, or “entry level”, was isolated for further analysis. This subset comprised 52 adverts within a broader sample of 150 UK job advertisements.

Each advert in the subset was evaluated across three repeated runs under identical conditions. Repeated evaluation was used to observe whether judgements remained stable, varied across runs, or converged toward more decisive classifications over time.

The focus of the analysis is not classification accuracy, but behavioural consistency under repeated execution. The objective is to examine how the system behaves when interpreting weak, mixed, or potentially ambiguous signals.

Findings should therefore be interpreted as observations of system behaviour under controlled evaluation conditions rather than as attributes of any single model.

4. Findings

4.1 Persistent “Unclear” Classification

Across all 52 early-career adverts, the system produced the same classification in every repeated run.

Each advert was classified as “Unclear”, with no movement into “Potentially Biased”, “Likely Biased”, or “Unbiased” across repeated evaluation. The system repeatedly returned the same indeterminate outcome despite identical re-evaluation conditions.

This pattern remained stable across all runs, with no evidence of threshold crossing or convergence toward more decisive classifications.

For example, adverts containing phrases such as “junior business analyst” or “graduate trainee” repeatedly produced “Unclear” classifications across all runs despite the consistent presence of early-career signals.

4.2 Absence of Variation

Unlike other subsets of the dataset, these adverts did not exhibit behavioural instability. There was no movement between adjacent categories and no evidence of drift, convergence, or escalation toward more decisive classifications across runs.

Judgement remained fixed in the same indeterminate position under repeated evaluation.

4.3 Confidence Behaviour

Confidence values remained tightly clustered across runs, typically centred around moderate levels. There was minimal variation in confidence and no instances of high-confidence classification or convergence toward stronger certainty.

This indicates that the system consistently expresses moderate confidence in an indeterminate outcome.

4.4 Contrast with Other Outputs

In the remaining adverts within the dataset, a different pattern was observed.

A proportion of cases exhibited variation across runs, particularly between “Potentially Biased” and “Unclear”. Some inputs produced more decisive classifications, and confidence values showed greater dispersion.

This contrast highlights a distinct behavioural difference. Some inputs produce instability at the boundary, while others remain persistently unresolved. Decision boundaries therefore produce more than one behavioural response.

Behavioural Responses at Decision Boundaries

Behavioural Pattern	Observed Behaviour	Example Outcome
Stable Resolution	Same definitive classification across repeated runs	Consistently “Unbiased”
Boundary Instability	Movement between adjacent classifications across runs	“Potentially Biased” ↔ “Unclear”
Non-Resolution	Persistent indeterminate classification across repeated runs	Repeated “Unclear” outcomes

These patterns indicate that decision boundaries can produce multiple behavioural responses rather than a single form of instability.

5. Interpretation

The observed pattern does not indicate that the system is failing to repeat its judgement. Instead, it indicates that the system is unable to move beyond an indeterminate classification despite repeated evaluation.

Where signals are weak, mixed, or open to interpretation, the system detects relevant cues but does not assign sufficient weight to cross the threshold required for decisive classification. As a result, the judgement remains fixed in an unresolved state.

This behaviour differs from instability.

In unstable cases, the system moves between adjacent categories across runs, reflecting sensitivity to marginal differences in interpretation. In non-resolving cases, the system does not move at all. It repeatedly returns the same indeterminate outcome despite detecting signals associated with potential bias.

These findings suggest that decision boundaries do not produce a single behavioural response. Some cases fluctuate under repeated evaluation, while others remain persistently unresolved.

The presence of non-resolution indicates that stability alone is insufficient for characterising system behaviour. A judgement may remain stable while still failing to provide a sufficiently decisive basis for action.

In operational settings, this distinction matters. A stable but unresolved output may appear reliable when viewed in isolation, despite not reaching a usable classification outcome.

6. Implications for AI Evaluation

These findings extend the interpretation of behavioural stability in AI systems.

First, stability should not be interpreted as evidence that a system has reached a definitive judgement. A stable output may represent persistent non-resolution rather than a reliable classification.

Second, evaluation approaches that focus on variation alone may overlook cases where the system consistently fails to commit. Non-resolution does not appear as instability, but it has important implications for how outputs are interpreted and used in practice.

Third, the presence of persistent “Unclear” outcomes suggests that AI systems may respond to weak or ambiguous signals by maintaining an intermediate position rather than converging toward a more decisive judgement.

In operational settings, this creates an important distinction between stable outputs and usable decisions. A system that repeatedly returns the same indeterminate classification may appear behaviourally consistent while still failing to provide a sufficient basis for action.

This is particularly relevant in contexts where unresolved cases require escalation, abstention, or human review rather than automated resolution.

Reliability therefore depends not only on whether a judgement remains consistent, but on whether it resolves into a usable form.

7. Limitations

This analysis focuses on a specific subset of recruitment text containing early-career signals. While the observed pattern is clear within this subset, further work is required to determine how widely non-resolution occurs across other domains and tasks.

The evaluation is conducted under controlled conditions using a fixed prompt and model configuration. Real-world use may introduce additional variability through changes in input structure or system settings.

The analysis does not assess classification accuracy. The purpose is to examine behavioural properties rather than determine whether individual outputs are correct.

8. Conclusion

When the same evaluative task is repeated, AI judgements do not always vary. In some cases, they remain stable without becoming decisive.

This paper identifies non-resolution as a distinct behavioural property of AI judgement systems. Rather than fluctuating between classifications, the system remains fixed in an indeterminate state, repeatedly returning the same unresolved outcome across repeated evaluation.

This behaviour differs from instability. The system does not change its judgement across runs, but nor does it move beyond ambiguity into a more definitive classification.

The findings show that decision boundaries do not produce a single behavioural response. Some cases fluctuate under repetition, while others remain persistently unresolved despite repeated evaluation under identical conditions.

This distinction is important for behavioural evaluation. A judgement may appear stable while still failing to provide a sufficiently usable basis for action. Stability alone therefore cannot be treated as evidence of reliable decision-making.

Understanding this distinction extends the evaluation of AI systems beyond individual outputs and beyond variation alone. Reliability depends not only on whether a judgement remains consistent, but on whether it resolves at all.

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